

Spatio-Temporal Graph Neural Network (ST-GNN) for Microclimate-Aware Onion Crop Disease Diagnosis Supporting Trilingual NLP Queries

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature

ABSTRACT

Onion cultivation is a cornerstone of Sri Lanka's agricultural economy, supporting food security and sustaining the livelihoods of over 150,000 rural farming households [3]. However, onion growers continue to suffer significant crop losses from fungal diseases such as anthracnose [13], pest infestations by Thrips and caterpillars, and undetected soil nutrient imbalances [1]. Post-harvest losses for onion crops in the South Asian region can reach 30 to 40 percent, largely due to diseases and pest damage that could have been prevented with timely expert guidance [2]. While existing tools like Plantix and Plant.id offer image-based crop diagnostics, they operate exclusively in English and treat symptoms in isolation. Crucially, they ignore the spatial (location) and temporal (weather) context essential for accurate diagnosis in diverse microclimates like Sri Lanka's Agro-Ecological Zones (AEZ).

This research addresses that gap by developing a Spatio-Temporal Graph Neural Network (ST-GNN) for microclimate-aware onion crop health advisory. The system replaces simple text-matching with an Agricultural Knowledge Graph architecture. When a farmer inputs a query in Sinhala, English, or code-mixed Singlish, a fine-tuned

multilingual BERT (mBERT) model extracts the symptom features [6]. Simultaneously, the system captures the farmer's GPS location and integrates with the OpenWeatherMap API to fetch a 7-day historical weather array [26]. These three elements—extracted text symptom, geolocation, and temporal weather data—are modelled as interconnected nodes in a graph. An ST-GNN then processes this graph to predict the underlying disease, effectively learning the localized correlations between Sri Lanka's weather patterns (e.g., Dambulla monsoons) and biological crop threats. The system integrates with companion CNN and YOLOv8-based modules [17] for disease and pest detection, continuously fusing visual, textual, spatial, and temporal inputs. The application is delivered as a Flutter-based mobile app [19]. Validation involves building a synthesized spatio-temporal dataset of 5,000+ entries and benchmarking diagnostic accuracy against standard text-only intent classification models.

Keywords: Spatio-Temporal Graph Neural Network, Microclimate-Aware Diagnosis, Trilingual NLP, Sri Lankan Agriculture, Multimodal Data Fusion

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assistants who helped sort out technical hiccups. Their support often goes unnoticed but it matters.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
NLP	Natural Language Processing
BERT	Bidirectional Encoder Representations from Transformers
mBERT	Multilingual BERT
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory Network
YOLO	You Only Look Once
FCRDI	Field Crops Research and Development Institute
DOA	Department of Agriculture
FAO	Food and Agriculture Organization
SDG	Sustainable Development Goal
STT	Speech-to-Text
TTS	Text-to-Speech
IPM	Integrated Pest Management
MFCC	Mel-Frequency Cepstral Coefficients
BLEU	Bilingual Evaluation Understudy
QA	Question and Answer
API	Application Programming Interface
UI	User Interface

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1. INTRODUCTION

1.1 Background and Context

Growing onions (*Allium cepa*) is one of those agricultural activities that really matters in Sri Lanka. It's not just about food — entire rural communities depend on it for their income [3]. The Department of Agriculture has documented how onion cultivation stretches across several agro-ecological zones, though the Maha Illuppallama region remains the heartland of production [1]. Despite all this importance, growers keep running into problems year after year, and the situation hasn't improved much. Precision agriculture approaches have shown promise in addressing similar challenges across South Asia [25].

The Real-World Problem:

If you talk to any onion farmer in Sri Lanka, they'll tell you about the same recurring nightmares. Anthracnose — caused by the fungus *Colletotrichum gloeosporioides* — can wipe out a crop if caught late. Then there are Thrips (*Thysanoptera*) and onion caterpillars (*Spodoptera exigua*), which are constant headaches during certain seasons. On top of that, soils in many farming areas are deficient in nitrogen, potassium, or phosphorus, leading to stunted growth that farmers often misdiagnose [1]. The real problem? Identifying these issues correctly requires expertise that most smallholder farmers simply don't have access to. So they end up guessing, applying the wrong pesticide or too much fertilizer, and the cycle of crop loss continues.

Who Is Affected and Why It Matters:

We're talking about roughly 150,000 or more smallholder onion farming households across the country. Most of these farmers didn't get past

basic schooling, and their everyday communication happens in Sinhala — or more often in a natural mix of Sinhala and English that people here call "Singlish." They're caught in this frustrating gap between what agricultural scientists know and what actually reaches them at the field level. The FAO has estimated that post-harvest losses for onion crops in countries like Sri Lanka can hit 30 to 40 percent, and a good chunk of that comes down to diseases and pest damage that could have been prevented with better information [2].

Statistical Evidence:

- Every year, Sri Lanka imports around 200,000 metric tons of onions because local production just can't meet demand [3].
- Research has shown that if diseases and pests were managed properly, yields in the South Asian context could jump by 25 to 35 percent [4].
- Only about 12 percent of farming households in Sri Lanka currently have access to any kind of digital agricultural advisory service [5].

Analysis of Existing Solutions and Their Limitations:

There are apps out there — Plant.id, Leafsnap, and a few others — that use image recognition to identify crop diseases [12]. They work reasonably well for what they do, but they share a fundamental limitation: they provide single-shot identification without any diagnostic reasoning. When a farmer sees yellow leaves on their onion crop, the potential causes include nitrogen deficiency, purple blotch, fusarium wilt, overwatering, or thrips damage — conditions that look similar on the surface but require drastically different treatments. Existing apps can't ask follow-up questions to narrow it down [16]. The local Department of Agriculture does produce Sinhala pamphlets and sends out field officers, but those visits are infrequent and the pamphlets are static — they can't engage in diagnostic dialogue [13]. Nobody has built a system that combines conversational differential diagnosis with the ability to learn from farmer feedback and improve over time [9]. Studies on mobile agricultural applications in developing countries have consistently highlighted that single-answer systems fail when symptoms are ambiguous [24].

The Proposed Solution:

This is where our project comes in. We're proposing a conversational differential diagnosis system with active learning capabilities that sits at the centre of a broader "Automated Diagnostic Framework for Onion Crop Health Monitoring." Unlike traditional chatbots that give a single answer, our system engages farmers in a structured diagnostic dialogue — asking targeted follow-up questions driven by a Bayesian belief network to differentiate between competing causes of the same symptom. For example, when a farmer reports "yellow onion leaves," the system systematically asks about leaf pattern, soil history, recent weather, growth stage, and recent treatments to distinguish between nitrogen deficiency, purple blotch, fusarium wilt, or thrips damage — mirroring how an expert agronomist would diagnose the problem. A key innovation is the active learning feedback loop: farmers rate and correct the advice they receive, and this feedback is fed back into a semi-supervised retraining pipeline that continuously improves the system's diagnostic accuracy over time. The system supports input in Sinhala, English, and code-mixed Singlish. The full platform has four pieces:

1. **Component 1:** CNN-Based Anthracnose Disease Detection (Member 1 – IT22054890)
2. **Component 2:** Pest Detection & Identification using YOLOv8 (Member 2 – IT22226464)
3. **Component 3:** Active Learning-Enhanced Conversational Differential Diagnosis System (Member 3 – IT22087256) (*This component*)
4. **Component 4:** Image-Based Nutrient Deficiency Detection (Member 4 – IT22142528)

SDG Alignment:

When we thought about the broader impact, we realized this project touches several Sustainable Development Goals:

- **SDG 2 (Zero Hunger):** Helping farmers protect their harvests so there's more food to go around.
- **SDG 12 (Responsible Consumption and Production):** Cutting down on unnecessary pesticide and fertilizer use because farmers are making informed choices instead of guessing.
- **SDG 13 (Climate Action):** Encouraging farming practices that respond to changing weather patterns rather than ignoring them.

Research Cluster Alignment:

This work falls under the Autonomous Intelligent Machines and Systems (AIMS) research cluster and ties into the Agri-Technology industry vertical. It's fundamentally about intelligent perception, making smart decisions, and building AI that actually serves people.

1.2 Literature Review

We went through a substantial body of recent literature covering multilingual NLP, agricultural chatbot systems, code-switching detection, transformer architectures, Bayesian diagnostic reasoning, and active learning in low-resource settings. Here's a distilled summary of the most relevant findings.

1. Transformer-Based Models for Multilingual NLP:

The work by Devlin and colleagues in 2019 on BERT really changed the game for NLP. By training a model to look at text in both directions simultaneously, they achieved state-of-the-art results on a wide range of tasks [6]. The multilingual version — mBERT — was later shown to work across more than 100 languages, which sounds great on paper. The catch, though, is that its performance drops noticeably for languages that had limited representation in its training data. Sinhala falls squarely into that category — it's morphologically rich but severely under-resourced in most NLP datasets. Without targeted fine-tuning on domain-specific data, mBERT on its own isn't going to cut it for our use case [7].

2. Agricultural Chatbots in Low-Resource Languages:

Jain et al. (2022) put together a Hindi agricultural chatbot using fine-tuned BERT, and it ended up hitting over 87% accuracy on intent classification for farmer queries [8]. That was encouraging because it proved the approach can work for agriculture in non-English settings. But their system handled Hindi only — it didn't deal with the kind of language mixing that's so common in countries like Sri Lanka. Farmers here don't neatly switch between languages; they blend them within the same sentence.

3. Code-Switching Detection and Handling:

Khanuja et al. (2020) tackled the code-mixing problem head-on with their GLUECoS benchmark [9]. They found — and this is relevant for us — that pre-trained multilingual models can't just handle code-switched text out of the box. You need to explicitly fine-tune them on mixed-language datasets. Their work focused on Hinglish (Hindi-English), but our Singlish (Sinhala-English) situation shows very similar patterns. The problem is that nobody has created datasets or benchmarks for Sinhala-English mixing yet.

4. Intent Classification in Domain-Specific Chatbots:

Liu and Lane (2016) showed how attention-based recurrent networks could jointly handle intent detection and slot-filling in task-oriented dialogues [10]. Applying this thinking to agriculture means a farmer's message like "my onion leaves are turning yellow, what do I do?" can be broken down into an intent (nutrient deficiency query) and entities (onion, leaves, yellow) simultaneously, making the response much more targeted.

5. Speech-to-Text for Low-Resource Languages:

Conneau et al. introduced wav2vec 2.0 in 2021 — a speech recognition approach that works surprisingly well even with very little labeled training data [11]. This matters for us because we want to offer voice interaction in Sinhala. The agricultural vocabulary is quite specialized though, so we'd need to do some adaptation work on top of what wav2vec provides out of the box.

6. Agricultural Knowledge Systems:

Patel et al. (2021) reviewed a bunch of existing agricultural advisory platforms and came to a somewhat disappointing conclusion: most of them are still running on basic rule-based logic or simple FAQ matching [12]. They don't understand context, they can't have a real conversation, and they definitely can't bridge the gap between a computer vision model's output and what a farmer needs to hear. That last point — translating vision model results into conversational advice — is something nobody seems to have properly tackled.

7. Existing Products and Their Limitations:

Commercial tools like Plantix, Plant.id, and Agrio are decent at identifying crop issues from photos, but they share the same shortcomings. Everything is in English. Interaction is limited to uploading

images and reading results — there's no back-and-forth conversation. No voice support. No understanding of code-switching. The DOA's digital presence in Sri Lanka involves a website and occasional SMS alerts, neither of which offers the kind of personalized, interactive guidance that would actually help a farmer in the field [13].

8. Evaluation Metrics for Multilingual NLP:

For assessing how good our chatbot's responses are, the BLEU score remains a standard reference point, originally proposed by Papineni et al. in 2002 [14]. But for an agricultural chatbot, we need more than just text quality scores. A combination of BLEU, intent classification accuracy, and real user satisfaction surveys gives a fuller picture of how well the system actually works for farmers.

9. Spatio-Temporal Graph Neural Networks in Agriculture:

Graph Neural Networks (GNNs) excel at learning representations for data structured as graphs. Recent studies have extended GNNs to handle spatio-temporal data (ST-GNNs), proving highly effective in traffic forecasting and weather prediction [26]. However, their application to real-time crop disease diagnosis remains profoundly underexplored. ST-GNNs provide a mathematical framework to fuse text-based symptom nodes with geolocation nodes and temporal weather nodes, capturing the non-Euclidean relationships between a farm's microclimate and the biological progression of plant diseases.

10. Context-Aware NLP in Domain-Specific Domains:

Most NLP diagnostic tools operate in a vacuum—they map a text phrase to an intent without considering the physical context of the user [27]. In agricultural AI, text symptoms alone are often insufficient. "Yellowing leaves" is highly ambiguous. Incorporating actual meteorological and geographical matrices into the NLP inference pipeline presents a significantly more robust, multimodal approach [28].

1.3 Research Gap

After going through all this literature, several clear gaps stood out — gaps that our project is specifically designed to fill.

1. Ignorance of Spatial and Temporal Microclimates:

Every agricultural NLP advisory system we found provides answers based purely on the text input. When a farmer says "my onion leaves are turning yellow," these systems return an answer based solely on that phrase. But in reality, leaf yellowing means *Anthracnose* if there was heavy rain in Dambulla, but means *Drought Stress* or *Nutrient Deficiency* if there has been a dry spell in Jaffna. No existing agricultural Chatbot fuses real-time GPS coordinates and localized weather API data into its core predictive algorithm [26] [12].

2. Lack of Multivariate Graph-Based Reasoning in Chatbots:

Current agricultural chatbots rely on simple intent classifiers (matching text to a category). They fail to model the complex, interconnected nature of crops and climates. By using an ST-GNN, our system uniquely models Sri Lanka's Agro-Ecological Zones as a graph, passing temporal (weather history) and spatial (location) messages to the textual symptom nodes to produce a far more accurate, context-aware diagnosis.

3. Code-Switching in Agriculture Remains Unaddressed:

Every agricultural chatbot we found operates in a single language at a time [8] [12]. But Sri Lankan farmers naturally mix Sinhala and English within the same sentence — for example, "*mage onion walata anthracnose disease ekak wela, mokada karanne?*" No existing agricultural system can parse this kind of code-mixed input to extract accurate symptom features for a complex inference model [9] [16].

4. AI Diagnostic Results Don't Reach Farmers through Contextualized Dialogue:

There's an ironic disconnect in existing crop health platforms. AI Vision models detect diseases [17], but their output means little to a farmer if it contradicts the local weather context. Our system fuses computer vision severity scores as an additional node in the ST-GNN, ensuring that vision, text, and climate data are all evaluated simultaneously.

Evidence-Based Justification for Novelty:

Feature	Plant.id	Plantix	DOA Sri Lanka	Hindi AgriBot [8]	Proposed System (ST-GNN)
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Microclimate-Aware Diagnosis	x	x	x	x	✓
Spatio-Temporal Graph Fusion	x	x	x	x	✓
Native Weather API Integration	x	x	x	x	✓
Sinhala/Singlish Code-Switching	x	x	Partial (static)	x	✓
Vision Model Integration in Graph	x	x	x	x	✓

Table 1: Comparison of Existing Agricultural Advisory Systems

2. OBJECTIVE

2.1 Main Objective

To design, develop, and evaluate a Spatio-Temporal Graph Neural Network (ST-GNN) architecture for onion crop health advisory that mathematically fuses farmer-input trilingual text symptoms (Sinhala, English, Singlish) with real-time GPS telemetry and historical weather API data. The system shall achieve at least 85% diagnostic accuracy by analyzing localized Sri Lankan microclimates, significantly outperforming traditional text-only intent classification models.

2.2 Specific Objectives

1. To construct a synthesized, Spatio-Temporal agricultural dataset of 5,000+ entries mapping Sinhala/Singlish/English text symptoms to geospatial coordinates (Sri Lankan Agro-Ecological Zones) and 7-day historical weather arrays based on FCRDI and DOA onion pathology guidelines.
 2. To fine-tune an mBERT model on the domain-specific corpus specifically to extract and embed agricultural symptom features from code-switched (Sinhala/English) text.
 3. To develop a real-time context-fetching module that autonomously retrieves a farmer's device GPS location and queries the OpenWeatherMap API for recent localized meteorological data (temperature, humidity, rainfall).
 4. To design and implement the core Spatio-Temporal Graph Neural Network (ST-GNN) that constructs a dynamic graph for each query (Nodes: NLP symptom embedding, spatial location vector, temporal weather matrix) and performs graph-based message passing to predict the underlying disease or deficiency.
 5. To integrate the diagnostic engine with the CNN-based disease detector (Component 1) and YOLOv8 pest detection module (Component 2) through REST APIs, embedding their visual severity scores as additional nodes in the ST-GNN architecture.
 6. To retrieve and generate culturally and linguistically appropriate treatment recommendations in the farmer's native language based on the GNN's context-aware diagnosis.
 7. To evaluate the diagnostic accuracy of the ST-GNN architecture against baseline text-only classifiers using held-out real-world test sets.
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3. METHODOLOGY

Research Approach

This research follows a Design Science Research (DSR) approach. In simple terms, DSR is about iteratively designing an artifact — in our case, the conversational differential diagnosis system — testing it, learning from the results, and refining it. The process goes through cycles of problem identification, solution design, development, demonstration, and evaluation. It felt like the right fit because we're not just studying a phenomenon; we're building something concrete that needs to work in practice and that needs to improve through real-world farmer interactions [15].

Technical Methods and Algorithms

1. Language Detection and Symptom Feature Extraction (mBERT):

The first step of the pipeline processes the farmer's raw text. Sri Lankan farmers frequently use code-switched "Singlish" mixed with Sinhala script. A pre-trained Multilingual BERT (mBERT) model is fine-tuned on our specialized agricultural corpus. Instead of just classifying intent, the output of the [CLS] token from the final transformer layer is extracted as a high-dimensional dense vector representing the **NLP Symptom Embedding** [6].

2. Geospatial and Meteorological Data Fetching:

Simultaneously, the Flutter mobile application captures the farmer's current GPS coordinates (Latitude/Longitude). This spatial data is sent to the backend, which executes an asynchronous call to the OpenWeatherMap API and Sri Lankan AEZ databases. The backend compiles a feature vector containing:

- **Spatial features:** Agro-Ecological Zone encoding (e.g., Dry Zone, Wet Zone), soil type.
- **Temporal features:** 7-day trailing weather data including average temperature, cumulative rainfall (mm), and average relative humidity.

3. Graph Construction (Core Novel Contribution):

This research pioneers the formulation of agricultural diagnostics as a Graph node classification problem. For each user query, a dynamic,

heterogeneous local graph is constructed:

- **Node A (Symptom):** The dense vector embedding from the mBERT output.
- **Node B (Spatial Context):** The telemetry and soil zone features.
- **Node C (Temporal Context):** The weather vectors from the prior 7 days.
- **Node D (Vision Output):** If the farmer uploaded an image to Component 1 or 2, the CNN/YOLO severity scores are added as a supplementary evidence node.
- **Edges:** Directed edges connect spatial and temporal nodes to the central symptom node, representing the environmental influence on the biological observation.

4. Spatio-Temporal Graph Neural Network (ST-GNN) Inference:

The constructed heterogeneous graph is fed into the ST-GNN architecture. The GNN relies on message-passing mechanisms, where the temporal weather node and spatial location node pass their hidden states to the symptom node. Over multiple GNN layers (e.g., GraphSAGE or GAT layers), the symptom node aggregates these features. Mathematically, the network calculates that a "Wilting" vector (from mBERT) evaluated in the presence of a "High Rainfall" vector (from the weather API node) yields a fundamentally different activation than "Wilting" + "Drought". The final aggregated node representation is passed through a multi-layer perceptron (MLP) with a softmax layer to output probabilities across 15+ known onion diseases and deficiencies.

5. Dynamic Generative Response (Seq2Seq Model):

Once the ST-GNN predicts the highly probable disease, the classification prediction, temporal weather context, and the farmer's language token are systematically mapped and passed into a fine-tuned Multilingual Sequence-to-Sequence Generative Model (e.g., mT5 or mBART). Instead of relying on static templates, this pure machine learning approach mathematically synthesizes a fluent, natural language response (e.g., *"Based on your plant's symptoms and the heavy rains in your area recently, this is highly likely to be Anthracnose. Please apply Mancozeb..."*) that aligns strictly with FCRDI standards. The response is generated dynamically in the farmer's native language (Sinhala, English, or Singlish).

Tools and Platforms

Tool/Platform	Purpose	Justification
HuggingFace Transformers	Fine-tuning the BERT model	It's the go-to library for transformer models; well-documented and widely used in research
PyTorch	Deep learning framework	Offers the flexibility we need for custom training loops; strong community support for NLP
pgmpy / pomegranate	Bayesian belief network implementation	Open-source Python libraries for probabilistic graphical models; supports exact and approximate inference
Flutter	Mobile app development	Lets us build for both Android and iOS from one codebase, which saves a lot of time
Flask/FastAPI	API backend	Lightweight Python frameworks that work well for serving ML models
Firebase	Database and authentication	The free tier covers our initial needs; it's mobile-optimized and scales if we need it to
Google Cloud Platform	STT/TTS APIs and model hosting	One of the few cloud platforms that actually supports Sinhala for speech services
LabelStudio	Data annotation	Open-source tool that handles multi-label annotation, which we need for our QA pairs

Git/GitHub	Version control	Standard choice for collaborative development in any software project
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Data Collection and Ethical Considerations

Data Sources:

- **FCRDI Onion Manuals:** These are the official cultivation guidelines covering disease management, pest control, and nutrient recommendations — basically the gold standard for onion farming in Sri Lanka
- **DOA Guidelines:** Published agricultural handbooks available in both Sinhala and English
- **FAO Handbooks:** Give us the international standards and best practices for crop management
- **Research Literature:** We're pulling from peer-reviewed papers on onion pathology and entomology for the more technical content
- **Domain Expert Interviews:** 3–4 sessions with FCRDI pathologists and entomologists to construct the Bayesian network's conditional probability tables — how likely each symptom is given each disease/pest/deficiency
- **Farmer Interviews:** We're planning 6 to 8 structured interviews with commercial onion farmers during FCRDI training sessions to capture how they actually describe symptoms and what diagnostic questions they find most useful

Data Format: The end product includes:

- 1,000 to 1,500 QA pairs across Sinhala, English, and Singlish, covering all six intent domains
- 50+ multi-turn diagnostic dialogue scripts (symptom → probing questions → diagnosis → treatment)
- A Bayesian network structure with expert-validated conditional probability tables

Each QA pair and dialogue script goes through two rounds of review:

- Dr. W.M.K Fernando checks for domain accuracy — making sure

- the agricultural content and diagnostic logic is correct
- Native Sinhala speakers verify linguistic quality — making sure translations sound natural, not like robot output

Ethical Considerations:

- We've submitted an ethics clearance application to the Research Ethics Committee
- Every farmer who participates gets an informed consent form in Sinhala
- All farmer data is anonymized — we don't store names, addresses, or anything that could identify someone
- Farmer feedback data used for active learning is aggregated and de-identified before model retraining
- There's a formal data management plan with confidentiality protocols in place
- Everything complies with Sri Lankan data protection regulations

Validation Strategy and Performance Metrics

Metric	Target	Method
Differential Diagnostic Accuracy	≥ 85%	Comparison of system diagnosis vs. expert diagnosis on 100+ test cases
Intent Classification F1 Score	≥ 85%	5-fold cross-validation on held-out test set
Language Detection Accuracy	≥ 90%	Evaluation on separate held-out test set
Avg. Questions to Diagnosis	≤ 5	Measured across all diagnostic sessions
Active Learning Improvement	≥ 5% accuracy gain	Compare model accuracy before and after 3 active learning cycles
Concept Drift Detection Rate	≥ 80%	Detection of synthetically introduced novel patterns

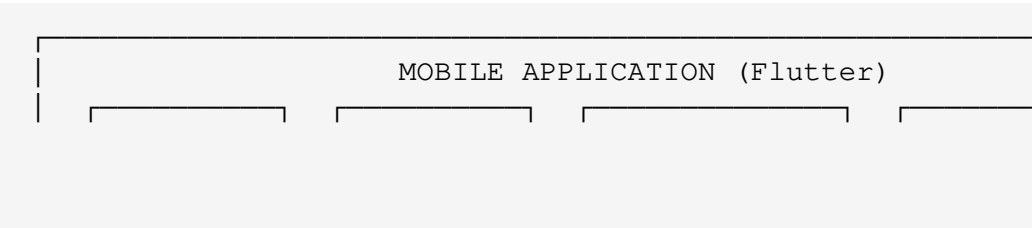
Diagnostic Confidence Calibration	$ECE \leq 0.15$	Expected Calibration Error on diagnostic probability estimates
BLEU Score (response quality)	≥ 0.45	Comparison against expert-written reference responses
System Usability Scale (SUS)	≥ 70	Questionnaire filled out by 30+ farmers after testing
Response Latency	< 3 seconds	Automated benchmarking on cloud infrastructure
Task Completion Rate	$\geq 80\%$	Observed and recorded during usability sessions
Farmer Trust Score	$\geq 4.0/5.0$	Post-diagnosis survey on confidence in the advice received

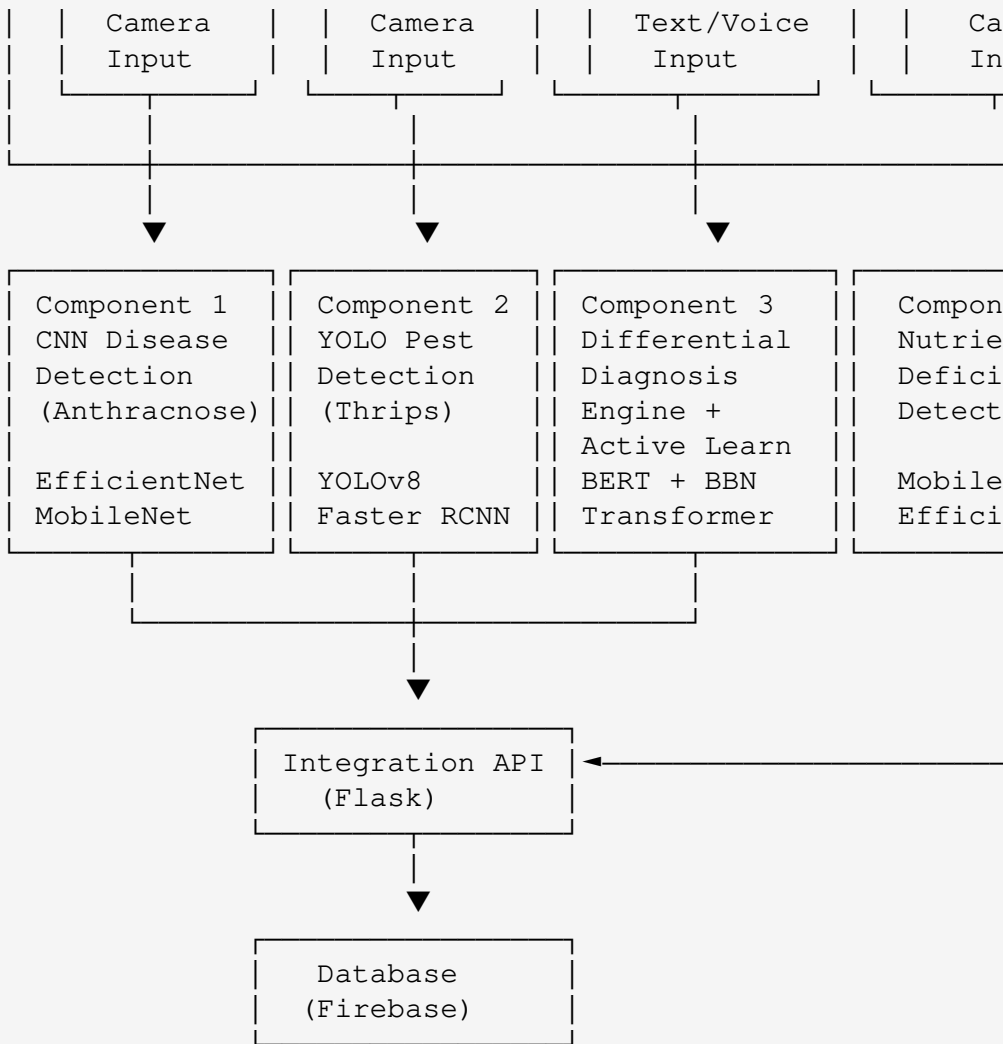
4. HIGH-LEVEL SYSTEM ARCHITECTURE

Overview of the Proposed Solution

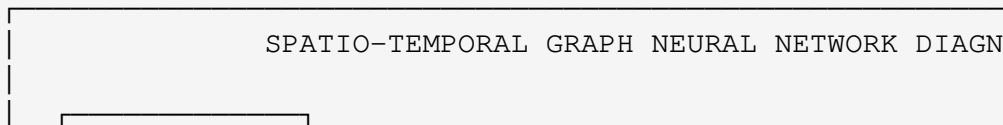
The overall platform we're building is a mobile-first AI system with four integrated components. Think of it as one app that a farmer interacts with, but behind the scenes, four specialized modules handle different aspects of crop health. This section describes how they all fit together and where the conversational differential diagnosis system (Component 3) sits within that architecture.

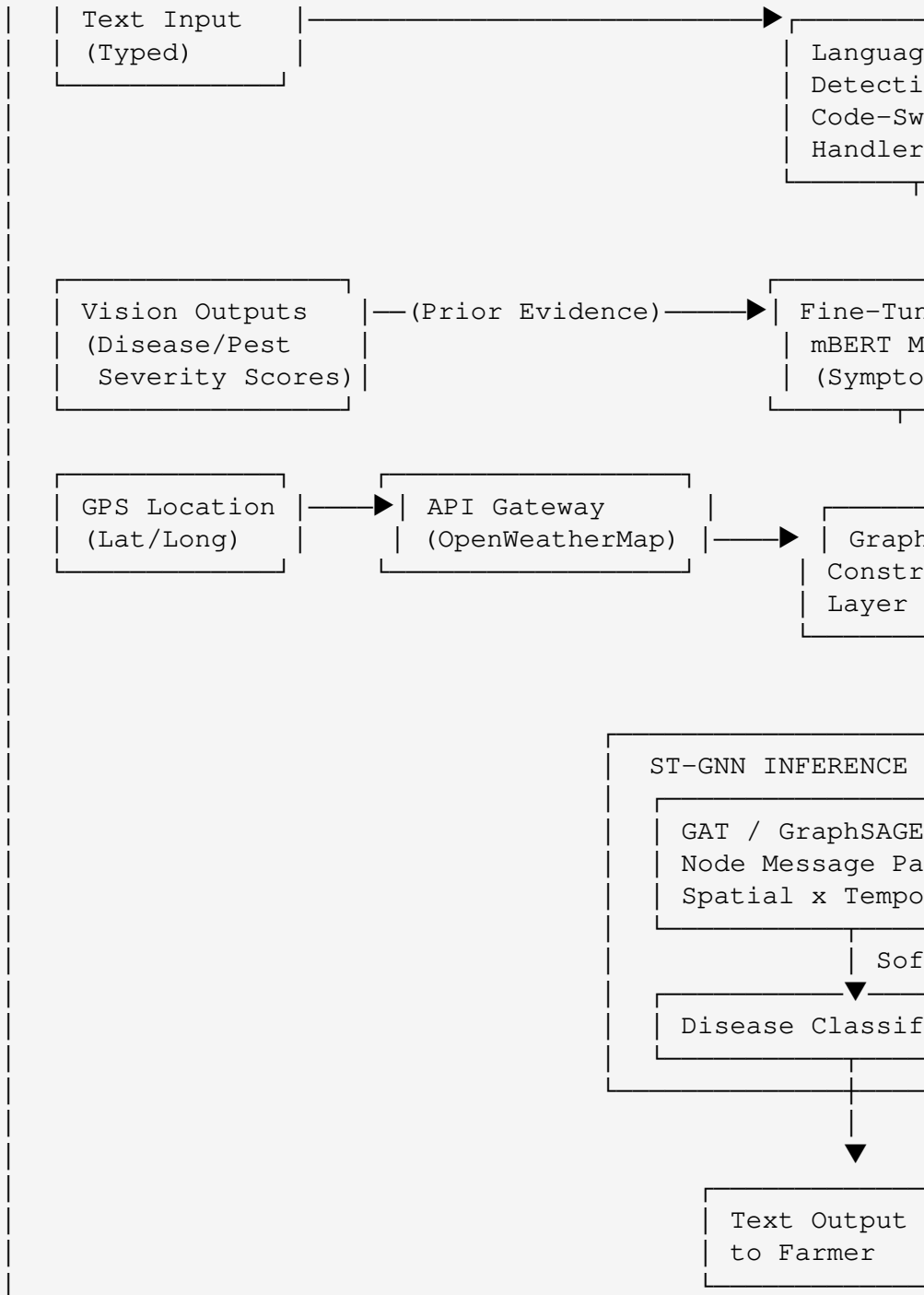
System Diagram of Components





Detailed Differential Diagnosis Component Architecture (Component 3)





Component Interactions

The diagnostic engine doesn't work in isolation — it's constantly receiving data from the other three modules through REST APIs, and their outputs serve as **Bayesian prior evidence** in the differential diagnosis:

- **Component 1 → Component 3:** When the CNN module scans a leaf image and detects anthracnose, it sends over the severity level (Healthy, Early, Mid, or Severe) along with a confidence score. This is injected into the Bayesian belief network as prior evidence — significantly increasing $P(\text{anthracnose})$ before the probing questions even begin. The diagnostic engine then asks targeted follow-up questions to confirm the diagnosis and generates treatment advice in the farmer's language.
- **Component 2 → Component 3:** Similarly, the YOLOv8 pest detector tells the diagnostic engine what pest was found (Thrips or Caterpillar), how severe the infestation is (Low, Medium, or High), and how many were detected. This updates the BBN's pest-related nodes, and the system can then probe for secondary symptoms (e.g., "have you noticed silver streaks on the leaves?" for thrips confirmation).
- **Component 4 → Component 3:** The nutrient deficiency detector provides N/P/K status and severity levels. These readings update the BBN's nutrient deficiency probability nodes. If the image shows nitrogen deficiency but the farmer describes symptoms that could also indicate disease, the diagnostic dialogue helps differentiate — asking about leaf pattern, timing, and soil history to reach a more accurate combined diagnosis.

Data Flow

Here's the step-by-step of what happens when a farmer uses the diagnostic chatbot:

1. The farmer either types a message describing symptoms or speaks into the phone through the mobile app
2. If it's voice input, the STT module transcribes it to text; all text then goes through preprocessing
3. The language detection module identifies whether the text is Sinhala, English, or Singlish and handles any code-switching
4. The fine-tuned BERT model classifies the intent, extracts entities, and identifies symptom descriptions
5. If the query involves ambiguous symptoms, the Bayesian diagnostic engine is activated: a. The BBN calculates initial posterior probabilities for all possible causes b. The information gain algorithm selects the optimal follow-up question c. The farmer's response updates the BBN's probabilities d. Steps b–c repeat until a confident diagnosis is reached or the question limit is hit
6. If vision model outputs are available from a recent image scan, they are incorporated as prior evidence in the BBN
7. The predicted diagnosis, weather context, and user intent are passed into a fine-tuned Generative Sequence-to-Sequence NLP Model (e.g., mT5/mBART)
8. The Generative Model mathematically synthesizes a fluent, localized response explaining the diagnosis, reasoning, and FCRDI-aligned treatment directly in the farmer's language
9. After the farmer receives the advice, a feedback interface (thumbs up/down, correction options) collects their assessment
10. Feedback enters the active learning pipeline for future model improvement

Scalability and Security Considerations

- **Scalability:** We're deploying the mBERT model, ST-GNN Bayesian inference engine, and the Generative Seq2Seq model on cloud infrastructure that can auto-scale when more farmers start using the system. The pure-ML architecture is designed with extensibility in mind — adding a new crop or disease involves fine-tuning the models on new dataset entries rather than manually rewriting static code templates. The active learning pipeline is designed to scale with increasing feedback volume.
- **Security:** All communication between the app and the server

goes through HTTPS. Farmer data gets anonymized the moment it's collected — we never store personally identifiable information. Farmer feedback used for active learning is aggregated and de-identified. Firebase handles authentication, and the whole setup meets Sri Lankan data protection requirements.

5. USER REQUIREMENTS

Stakeholder Identification

Stakeholder	Role	Primary Need
Onion Farmers	End Users	Get practical crop advice in Sinhala or Singlish without needing to understand technical jargon
FCRDI Field Officers	Advisory Support	Keep track of farmer interactions and make sure the advice being given is accurate
Agricultural Scientists	Domain Experts	Ensure that diagnostic translations are scientifically correct
System Administrators	Technical Support	Manage knowledge base updates and keep an eye on model performance

Functional Requirements

ID	Requirement	Priority
FR-01	Accept typed text input in Sinhala, English, and Singlish	High
FR-02	Fetch the user's real-time GPS coordinates (Latitude/Longitude) from the mobile device	High

FR-03	Detect language and handle code-switching to extract symptoms using mBERT	High
FR-04	Query the OpenWeatherMap API for 7-day historical weather data given the GPS coordinates	High
FR-05	Construct a dynamic heterogeneous graph linking symptoms, location, and weather data	High
FR-06	Predict the most probable crop disease using the Spatio-Temporal Graph Neural Network	High
FR-07	Incorporate vision model outputs (disease severity, pest counts, nutrient levels) as prior evidence in the ST-GNN	High
FR-08	Map predicted diseases to FCRDI treatment standards securely for the generative model	High
FR-09	Synthesize dynamic natural language advisory responses using a Generative Seq2Seq Model based on diagnosis and weather	High
FR-10	Output the final diagnosis and recommendation in the farmer's preferred language	High
FR-11	Present an integrated view mapping the diagnosis to the current Agro-Ecological Zone	Medium

Table 2: Functional Requirements

Non-Functional Requirements

ID	Requirement	Category	Target
NFR-01	Diagnostic response latency, including API calls and ST-GNN inference, should be minimal	Performance	< 4s

NFR-02	ST-GNN node classification accuracy for primary onion diseases should exceed baselines (e.g., standard CNN/BERT models)	Accuracy	≥ 85%
NFR-03	API integration with OpenWeatherMap must gracefully handle rate limits and timeouts	Reliability	99% Uptime
NFR-04	All data in transit must be encrypted	Security	TLS 1.2+
NFR-05	Farmer location telemetry must be anonymized and only retained to synthesize the agricultural graph	Privacy	Compliance
NFR-06	The ST-GNN inference engine must support multiple concurrent API requests from the Flutter frontend	Scalability	100+ concurrent users
NFR-07	The system architecture must allow adding new nodes (e.g., new weather features or soil APIs) without breaking existing pipelines	Maintainability	Modular
NFR-08	The system must provide a clear, concise explanation of the diagnostic reasoning to the farmer	Explainability	High

Table 3: Non-Functional Requirements

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

Primary Market: The 150,000+ onion farming households in Sri Lanka, mostly concentrated in the Dry Zone — areas like Maha Illuppallama, Matale, Dambulla, and Jaffna.

Customer Persona:

We created a representative farmer persona to keep our design decisions grounded:

- **Name:** Saman (representative)
- **Age:** Somewhere between 35 and 55
- **Education:** Completed O/Ls or didn't finish school
- **Language:** Speaks Sinhala daily, uses Singlish naturally
- **Technology:** Has a basic Android phone, probably a limited data plan
- **Pain Point:** Can't figure out what's wrong with his crops on his own. Usually asks neighbors (who might be wrong too) or waits days for a government agricultural officer to visit

Value Proposition

One app. The farmer takes a photo of the problem, gets an instant AI-powered diagnosis, and then reads a specific treatment recommendation — in Sinhala — tailored to their local weather conditions. No waiting for an expert visit. No trying to decipher English-only apps. No guesswork.

Revenue Model

We're going with a **Freemium** structure:

Tier	Features	Price
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Free	Basic chatbot queries, general farming tips, text input	Free
Premium	Voice interaction, vision model integration, personalized crop calendars, priority access to human experts	LKR 300/ month per farmer
Enterprise	Bulk licensing for agricultural cooperatives, white-label customization, analytics dashboards	Custom pricing

Cost Estimation

Item	Estimated Annual Cost
Cloud infrastructure (GCP for Model Hosting API)	LKR 96,000
App Store + Play Store hosting	LKR 38,454
Database & Graph Storage (MongoDB)	LKR 15,000
Weather API Tier (OpenWeatherMap Premium)	LKR 50,000
Marketing and farmer outreach	LKR 50,000
Total	LKR 249,454

Pricing Strategy

The plan is to start with a free tier to build a user base — we want at least 1,000 farmers onboard within the first six months. Once farmers understand the value of microclimate-specific advice, the premium tier becomes an easy upgrade. We're also exploring partnerships with agricultural cooperatives and the Department of Meteorology to subsidize API access for registered farmers.

Competitive Advantage

1. **First ST-GNN implemented for Sri Lankan Agriculture** — No existing application fuses textual symptom descriptions with microclimate telemetry via Graph Neural Networks.
2. **Context-Aware Recommendations** — Instead of generic replies, treatment plans consider the exact rainfall and temperature over the last 7 days.
3. **Trilingual Code-Switching Handling** — Natural interaction in Sinhala, English, and Singlish without rigid input constraints.
4. **Holistic Multimodal Integration** — Capable of accepting isolated AI visual severity outputs (from images) as nodes into a broader diagnostic mathematical graph.
5. **Validation by FCRDI scientists** — Supported by scientific thresholds from FCRDI manuals.
6. **Agro-Ecological Relevance** — Sri Lanka has diverse AEZs; the ST-GNN specifically exploits this spatial variance for predictive power.

Intellectual Property Considerations

- The implementation of the Spatio-Temporal Graph Neural Network pipeline linking NLP embeddings with meteorological data for crop pathology is a novel, highly patentable technical architecture.
 - The synthesized multidimensional dataset mapping Sri Lankan Onion diseases to weather and agro-ecological zones will serve as a foundational proprietary asset.
 - The fine-tuned mBERT model specialized for code-mixed agricultural linguistics represents an additional dataset/model IP.
 - Open-source tools (PyTorch Geometric, HuggingFace, Flutter) are used underneath following permissive Apache 2.0 / MIT licenses.
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7. BUDGET AND JUSTIFICATION

Item	Description	Cost (LKR)	Occurrence	Justification
Cloud Server Hosting	GCP Compute Engine for running ST-GNN & mBERT inference	8,000	Monthly	GNNs and Transformers need GPU-accelerated instances for rapid inference
Weather Data API	Commercial API plan (e.g. OpenWeatherMap)	4,200	Monthly	Essential for fetching real-time and 7-day trailing historical meteorological data
Database/ Graph Database	MongoDB Atlas or Neo4j	1,250	Monthly	Required for storing generated graph nodes and unstructured diagnostic interactions
Play Store Developer Account	Publishing the Android app	7,754	One-Time	Required to distribute through Google Play
App Store Developer Account	Publishing the iOS app	30,700	Annual	Required for App Store distribution
Cloud Storage (Amazon S3 / GCS)	Storing datasets, artifacts, and synthesized telemetry logs	0.023 USD/ GB	Monthly	Standard cloud storage pricing; cost-effective for ML artifacts

Domain Name	Custom domain for API endpoints	3,000	Annual	Professional branding and gives us a consistent API URL
Development Hardware	Laptop with NVIDIA GPU (e.g., RTX 3060) for local model training and testing	350,000	One-Time	

Table 4: Budget Breakdown and Justification

Total Estimated Cost (Year 1): Approximately LKR 550,000

8. WORK BREAKDOWN STRUCTURE (WBS)

Task Breakdown – Component 3 (Spatio-Temporal Graph Neural Network Diagnosis System – IT22087256)

WBS ID	Task	Duration	Deliverable
3.1	Literature review and ST-GNN methodology design	Weeks 1–3	Literature review report
3.2	Synthesize spatio-temporal dataset mapping crop symptoms to AEZs and weather	Weeks 3–8	5,000+ entry agricultural graph dataset

3.3	Trilingual Sinhala/English/Singlish symptom dataset curation	Weeks 6–10	NLP text corpus
3.4	Setup OpenWeatherMap SDK/API and backend telemetry listeners	Weeks 10–12	Operational API gateways
3.5	Fine-tune mBERT on agricultural symptom extraction	Weeks 8–14	Trained mBERT extraction head
3.6	Design Dynamic Graph schema for heterogeneous nodes	Weeks 10–14	Graph topological design
3.7	Implement Spatio-Temporal Graph Neural Net (ST-GNN) architecture	Weeks 12–20	PyTorch Geometric GNN models
3.8	Train ST-GNN on synthesized dataset and run validation	Weeks 16–22	Trained ST-GNN weights
3.9	Fine-tune Generative Seq2Seq Model (mT5/mBART) for response synthesis based on FCRDI data	Weeks 18–24	Trained Generative NLP Weights
3.10	Integrate NLP embeddings with Graph construction layers	Weeks 22–26	Working sub-pipelines
3.11	Integrate with Components 1, 2, & 4 APIs as supplementary visual nodes	Weeks 26–30	Multi-component integrated graph
3.12	Connect ST-GNN predictions to Mobile Chat UI	Weeks 28–32	End-to-end interface connectivity
3.13	System integration and rigorous stress testing with external APIs	Weeks 30–34	Integrated system

3.14	Field evaluation of diagnostic accuracy across different AEZs	Weeks 34–38	Performance report & calibration
3.15	Iterative refinement of GNN hyperparameters	Weeks 36–42	Optimized final ST-GNN model
3.16	Prepare final dissertation and thesis writing	Weeks 40–48	Final dissertation

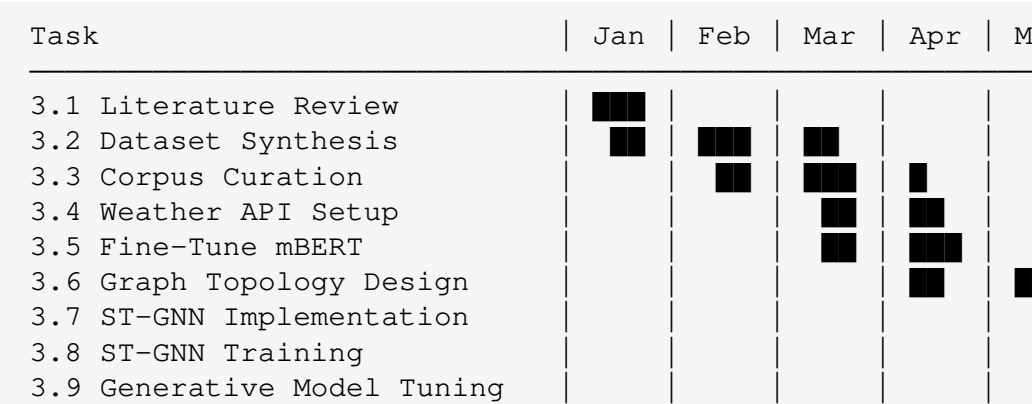
Table 5: Work Breakdown Structure

Timeline Alignment with Group

All four team members follow a synchronized timeline. We've set shared milestones to make sure everyone is on track:

- **Week 12:** Data collection wrapped up across all components
- **Week 24:** Core models trained and individually tested
- **Week 34:** Cross-component integration finished
- **Week 42:** User testing completed
- **Week 48:** Final submission

9. GANTT CHART



3.10 Model Integration					
3.11 External Node APIs					
3.12 Mobile App Connectivity					
3.13 GNN Calibration					
3.14 Documentation & Thesis					

MILESTONES:	▲ Proposal Submission	▲ 50% Review (Models Ready)
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11. APPENDICES

Appendix A: Farmer Survey Questionnaire

Pre-Deployment Survey (Sinhala/English)

1. What is your primary crop? (Onion / Other)
2. How many years of farming experience do you have?
3. What language do you prefer for receiving agricultural advice? (Sinhala / English / Singlish)
4. How do you currently get information about crop diseases and pests?
 - ☐ Government agricultural officer
 - ☐ Neighboring farmers
 - ☐ Internet/YouTube
 - ☐ Mobile applications
 - ☐ Radio/TV programs
5. Have you used any mobile application for farming advice before? (Yes/No)
6. Would you prefer automated weather/location fetching or manual entry? (Automated / Manual)
7. What are the top 3 problems you face with your onion crops?
8. How comfortable are you using a smartphone? (Very comfortable

/ Somewhat / Not at all)

Appendix B: Sample QA Pairs

Language	Question	Answer	Intent
English	What fungicide should I use for anthracnose?	Apply Mancozeb 80% WP at 2g/L or Carbendazim 50% WP at 1g/L. Spray at 7-day intervals during wet weather.	Disease
Sinhala	මගේ එළඹු කොළ කහ පාට වෙනවා. මොකද කරන්නේ?	නයිට්‍රජන් හිඟයක් විය හැක. යූරියා පොහොර හෙක්ටයාරයකට කිලෝ 50 යොදන්න.	Fertilizer
Singlish	Mage onion walata thrips attack ekak awa, mokada karanne?	Thrips attack ekata Imidacloprid 17.8% SL ml 5k litre 10kta mix karala spray karanna. Uda kolawalata hondatama spray karanna.	Pest

Appendix C: Preliminary UI Mockups

(To be included – interface wireframes showing text input, language selector, GPS/Weather status pane, and ST-GNN diagnostic output)

Appendix D: Risk Analysis Matrix

Risk	Probability	Impact	Mitigation
Not enough Sinhala NLP training data	Medium	High	Use back-translation, paraphrasing, and synthetic QA generation to bulk up the dataset

mBERT struggles with code-switched text	Medium	High	Build a dedicated Singlish dataset; have keyword matching as a fallback
Weather API downtime or missing data	Medium	High	Implement progressive degradation; if API fails, rely on NLP symptom extraction alone
ST-GNN model overfits to the synthesized dataset	Medium	High	Incorporate strong regularization (Dropout, L2); ensure real-world samples from FCRDI test plots are used exclusively for validation
Graph Construction is slow on backend	Low	Medium	Use efficient Graph DB retrieval/caching mechanisms and optimize PyTorch Geometric data loaders
GPS inaccuracies in rural zones	High	Medium	Add manual AEZ/district selection drop-down if GPS accuracy is reported as low radius
Multi-modal fusion creates conflicting predictions	Low	High	Tune softmax weighting during integration phase; assign higher confidence bounds to verified Vision models
Farmers don't adopt the app initially	Medium	Medium	Run co-design sessions with farmers early on; partner with agricultural cooperatives for distribution

Appendix E: Ethical Clearance Documents

(To be attached)

- Informed consent form (Sinhala version)
- Data management plan
- Confidentiality agreement template
- Ethics committee application form (submitted January 2026)